PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

On

TravelBuddy(A Shared Travel Coordination Platform)

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted By:

1. Ashish Mehra (2211981110)
2. Arnab Pramanik(2211981091)
3. Arup Maiti (2211981096)
4. Ayush Jagota (2211981120)

Supervised By:

Dr. Monika Aggarwal (9463865180)
Assistant Professor
Chitkara University, Rajpura

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY

CHITKARA UNIVERSITY, PUNJAB, INDIA

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **TravelBuddy (A Shared Travel Coordination Platform)** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Monika Aggarwal.** The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura
Date: 14-05-2026

Ashish Mehra
University Roll No: 2211981110

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Dr. Monika Aggarwal
Assistant Professor
Department of Computer Science and Engineering
Chitkara University Institute of Engineering and Technology,
Chitkara University, Punjab, India

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I sincerely acknowledge and appreciate everyone who contributed directly or indirectly to the successful completion of this project. Their support and encouragement played a significant role in achieving this milestone.

ABSTRACT

Travel planning has become increasingly popular among students, professionals, and tourists. People often use online platforms and mobile applications to search for destinations, create travel plans, and manage trips. However, many existing travel platforms mainly focus on booking hotels or transportation and do not provide proper support for connecting travelers who are interested in visiting the same destination. As a result, many travelers face difficulties in finding travel companions, managing trip details, and coordinating travel activities effectively. Solo travel can also increase travel expenses and may reduce the overall travel experience for many users.

The objective of this project is to develop a web-based travel planning and travel companion platform called **TravelBuddy**, which helps travelers plan trips and connect with other travelers easily. The system integrates multiple travel-related features into a single platform so that users can create trips, join travel groups, communicate with travelers, manage travel requests, and share expenses without using multiple applications. By combining these features, the platform aims to make travel planning more organized, convenient, and cost-effective.

TravelBuddy is developed as a web application using the **MERN stack**, which includes **MongoDB, Express.js, React.js, and Node.js**. The frontend interface is built using **React.js** and styled with **Tailwind CSS** to provide a responsive and user-friendly design. The backend is implemented using **Node.js** and **Express.js** to manage API requests and application logic, while **MongoDB** is used for storing user information, trip details, chat messages, travel requests, and notifications. The platform also integrates **Socket.io** for real-time communication between travelers and supports multiple pickup points for convenient trip joining.

Users can register and securely log in to the platform using **JWT authentication**. After logging in, users can create travel plans, explore trips created by others, send trip requests, communicate through real-time chat, and receive notifications and travel updates instantly. The system also supports expense sharing features, helping travelers reduce travel costs by sharing trip expenses and resources. A dashboard is provided to help users manage trips, track activities, and view travel-related updates in an organized manner.

Overall, the TravelBuddy platform is designed to improve the travel experience by simplifying trip planning and helping travelers connect with suitable travel companions. By integrating multiple travel management features into one platform, TravelBuddy provides a more organized, interactive, and user-friendly solution for modern travelers while also helping reduce travel costs and improve travel convenience.

1. Introduction to Project

1.1 Project Background

In recent years, online travel platforms and digital services have become an important part of modern travel planning. Travelers widely use websites and mobile applications to search for destinations, book transportation, manage trips, and explore travel experiences. These digital platforms provide convenience, easy access to travel information, and better communication opportunities for travelers. However, despite the availability of many travel applications, travelers still face difficulties in planning trips efficiently and connecting with suitable travel companions.

Planning trips individually can often be time-consuming and expensive. Travelers usually need to use multiple applications for trip planning, communication, expense management, and travel updates. This makes the travel process fragmented, less organized, and difficult to manage effectively. In many cases, travelers also struggle to find people who share similar travel interests, destinations, or schedules. Solo travel may increase expenses and sometimes reduce safety, comfort, and overall enjoyment during the journey.

With the rapid growth of web technologies and real-time communication systems, new opportunities have emerged to improve travel management platforms. Modern technologies can help travelers connect instantly, share trip information, communicate in real time, and manage travel activities more efficiently. Integrating these technologies into a single travel platform can significantly improve the overall travel experience and make trip planning more convenient, interactive, and cost-effective.

1.2 Problem Statement

Despite the availability of online travel platforms and digital travel services, travelers still face several challenges while planning and managing trips. One of the major issues is that organizing travel plans, finding suitable travel companions, and managing trip details often require significant time and effort. Travelers frequently struggle to coordinate destinations, schedules, budgets, and communication efficiently.

Another challenge is the lack of integrated travel management features. Most existing platforms mainly focus on hotel or ticket booking and provide limited support for traveler interaction, real-time communication, expense sharing, or trip coordination. As a result, travelers often need to use multiple applications for trip planning, chatting, notifications, and managing travel expenses. Switching between different platforms makes the travel process less organized and reduces convenience.

Furthermore, solo travelers often face difficulties finding people with similar travel interests or destinations. Traveling alone can increase travel expenses and may sometimes reduce safety, comfort, and overall enjoyment during the journey. These limitations highlight the need for a unified system that integrates trip planning, traveler connection, communication, notifications, and expense sharing into a single platform.

1.3 Objective of the Project

The main objective of the **TravelBuddy** project is to develop a web-based travel planning and travel companion platform that helps travelers plan trips more efficiently and connect with other travelers in a more organized manner.

The specific objectives of the project are:

- To allow users to create, manage, and explore travel plans within a single platform.
- To help travelers connect with people having similar destinations and travel interests.
- To provide secure user authentication and profile management.
- To enable real-time communication between travelers using chat features.
- To support multiple pickup points for convenient trip coordination.
- To provide trip request management for joining and organizing trips.
- To help users reduce travel costs through expense sharing features.
- To provide notifications and travel updates for better trip management.
- To provide a dashboard for tracking user trips, activities, and travel information.

By achieving these objectives, **TravelBuddy** aims to simplify travel planning, improve traveler interaction, and provide a more convenient, affordable, and enjoyable travel experience.

The platform also helps travelers connect with like-minded people, making journeys safer, more social, and better organized. By integrating multiple travel features into one system, TravelBuddy reduces the need for using separate applications and improves overall travel management efficiency.

1.4 System Overview

TravelBuddy is designed as a web-based travel planning and travel companion platform that combines trip management, traveler interaction, and real-time communication to create an organized and user-friendly travel environment. The system is implemented using the **MERN stack**, which includes **MongoDB, Express.js, React.js, and Node.js**. Secure user authentication is implemented using **JSON Web Tokens (JWT)** to ensure safe and authorized access to the platform.

After logging into the system, users can create and manage their travel plans easily. The platform allows users to explore trips created by other travelers and connect with people who have similar travel interests and destinations. Travelers can send trip requests, join travel groups, and coordinate trips through the platform.

The system integrates **Socket.io** for real-time communication, enabling users to chat instantly and receive live travel updates and notifications. Multiple pickup point support is also provided to help travelers join trips more conveniently from nearby locations.

TravelBuddy also includes features such as expense sharing, allowing travelers to divide trip costs and manage travel expenses efficiently. All user information, trip details, chat messages, notifications, and travel requests are stored securely in the **MongoDB database**.

The platform also provides a dashboard that displays travel activities such as created trips, joined trips, notifications, and chat updates. Additional features such as trip management and real-time updates help users organize their journeys in a more efficient and convenient manner.

TravelBuddy is developed to provide a smart and organized solution for modern travel planning and traveler interaction. The platform allows users to create travel plans, discover trips, connect with travelers, and communicate in real time within a single system. By integrating trip management and traveler coordination features, the platform helps users avoid the difficulties of using multiple applications for planning and communication. The system is built using the MERN stack, including MongoDB, Express.js, React.js, and Node.js, while JWT authentication ensures secure access and protection of user data.

The platform also supports real-time chat, travel notifications, expense sharing, and multiple pickup points to improve travel convenience and coordination. Users can manage trips, send requests, join travel groups, and track activities through a user-friendly dashboard. TravelBuddy aims to make traveling more affordable, interactive, and efficient by helping travelers connect with people who share similar destinations and travel interests.

1.5 Expected Benefits

The **TravelBuddy** platform is designed to improve the travel planning experience by integrating multiple travel-related features into a single system. By combining trip management, traveler connection, real-time communication, and expense sharing within one platform, the system reduces the need to use multiple applications for travel coordination and management.

The platform helps travelers save time while planning trips, finding travel companions, and managing travel activities. Features such as real-time chat, travel notifications, and multiple pickup points improve communication and coordination between travelers. Expense sharing features also help users reduce travel costs and make group travel more affordable and convenient.

The dashboard helps users stay organized by providing easy access to trip details, notifications, joined trips, and travel activities. Secure authentication and efficient trip management improve the overall user experience and ensure safe access to travel information.

Overall, TravelBuddy provides a modern web-based solution for travel planning and traveler interaction, making travel more organized, social, affordable, and enjoyable for users.

2. Methodology

The **TravelBuddy** system is designed using a modular architecture where different components work together to provide efficient travel planning and traveler interaction services. The architecture divides the system into multiple modules such as authentication, trip management, traveler connection, real-time communication, notifications, and expense sharing. Each module performs a specific task, which helps improve system efficiency, scalability, and maintainability.

The modular design also makes it easier to update or improve individual components without affecting the entire system. This approach ensures that the TravelBuddy platform operates in a structured and organized manner while providing reliable travel planning, communication, and trip management features to users.

The system follows a client-server architecture where the frontend handles user interaction and the backend manages business logic, database operations, and real-time communication. The frontend is developed using React.js and Tailwind CSS to provide a responsive and user-friendly interface, while the backend is implemented using Node.js and Express.js. MongoDB is used for storing user information, trip details, chat messages, and notifications securely. Socket.io is integrated to support real-time chat and live travel updates between travelers.

2.1 System Approach

The system follows a web-based architecture where the frontend interacts with the backend server through APIs. The backend processes requests, manages user data, and handles real-time communication between travelers. The platform is developed using the MERN stack, which provides a scalable and efficient environment for managing travel-related activities and user interactions.

The frontend is responsible for displaying travel information, trip details, chat interfaces, and dashboard activities to users. The backend handles authentication, trip management, notifications, expense sharing, and database operations. MongoDB is used to securely store user information, trip records, chat messages, and travel requests, while Socket.io enables real-time communication and live updates between users.

The system is designed to provide a smooth and organized travel experience by integrating all essential travel management features into a single platform. Each module communicates with the backend server through secure APIs, ensuring proper data handling and efficient system performance. The modular structure of the platform also improves maintainability and allows future enhancements such as AI-based travel recommendations, map integration, and mobile application support without affecting the existing system functionality.

The major modules of the system include:

- User Authentication Module
- Trip Creation and Management Module
- Traveler Connection Module
- Real-Time Chat and Communication Module
- Trip Request Management Module
- Expense Sharing Module
- Notification and Travel Updates Module
- Dashboard and Activity Tracking Module

2.2 Tools and Technologies

The following technologies were used to develop the TravelBuddy platform.

- **Frontend:** React.js is used to build a dynamic and responsive user interface.
- **Backend:** Node.js and Express.js are used to handle server operations and API development.
- **Database:** MongoDB is used to store user data, uploaded documents, flashcards, and quiz results.
- **Artificial Intelligence:** Google Gemini API is used to analyze document content and generate summaries, explanations, flashcards, and quizzes.
- **Authentication:** JSON Web Tokens (JWT) are used to provide secure login and user authentication.
- **User Interface:** Tailwind CSS is used to design a modern and responsive interface.
- **Development Environment:** Visual Studio Code is used as the primary development tool.

2.3 Flow of the System

The workflow of the **TravelBuddy** system is as follows:

1. User registers and logs into the system.
2. The user creates or explores travel plans on the platform.
3. The system stores and manages trip details and traveler information.
4. Users can send trip requests and connect with travelers having similar destinations and interests.
5. The system manages notifications, expense sharing, and multiple pickup points for trips.
6. The user can view trips, chat activity, notifications, and updates through the dashboard.

3. Design

3.1) Use Case Diagram:

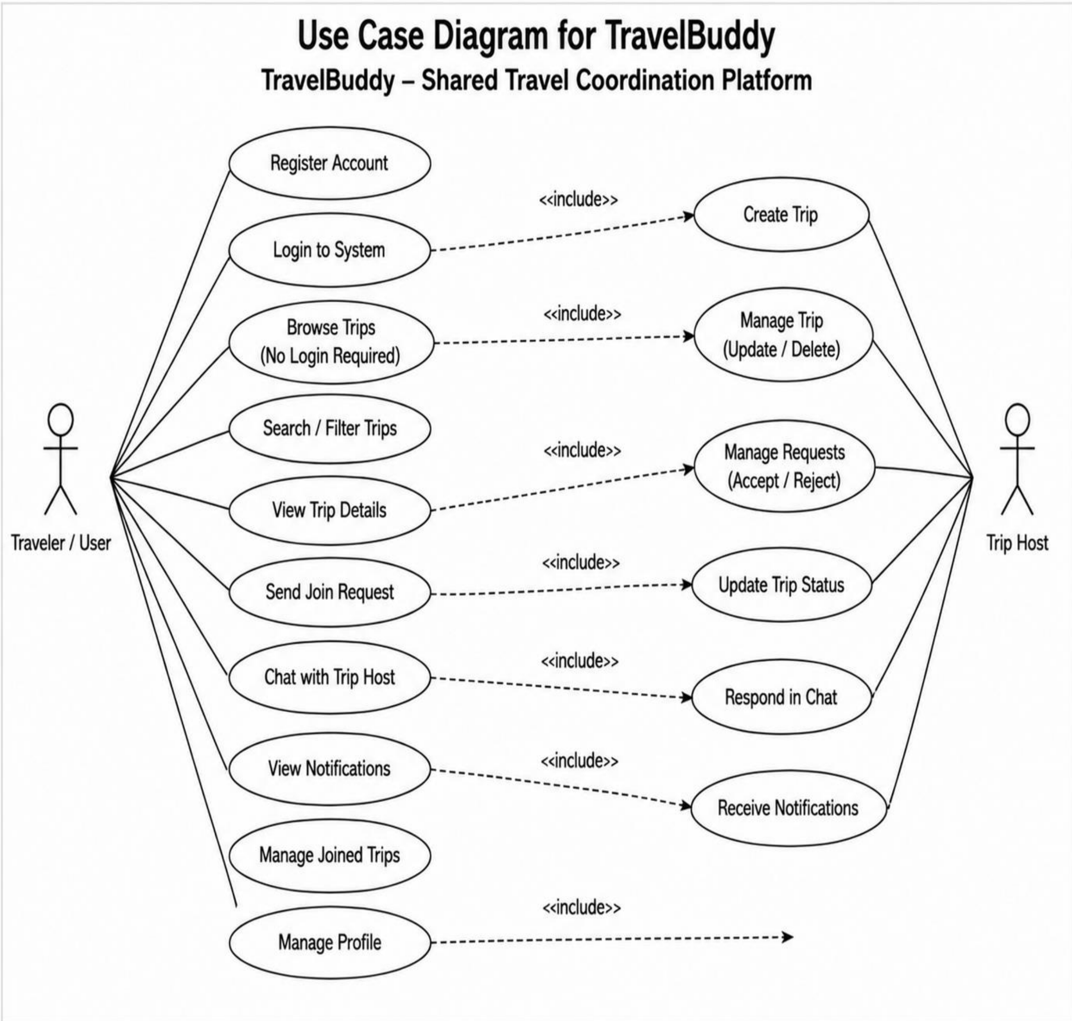

Fig: 3.1

3.2) System Architecture Diagram:

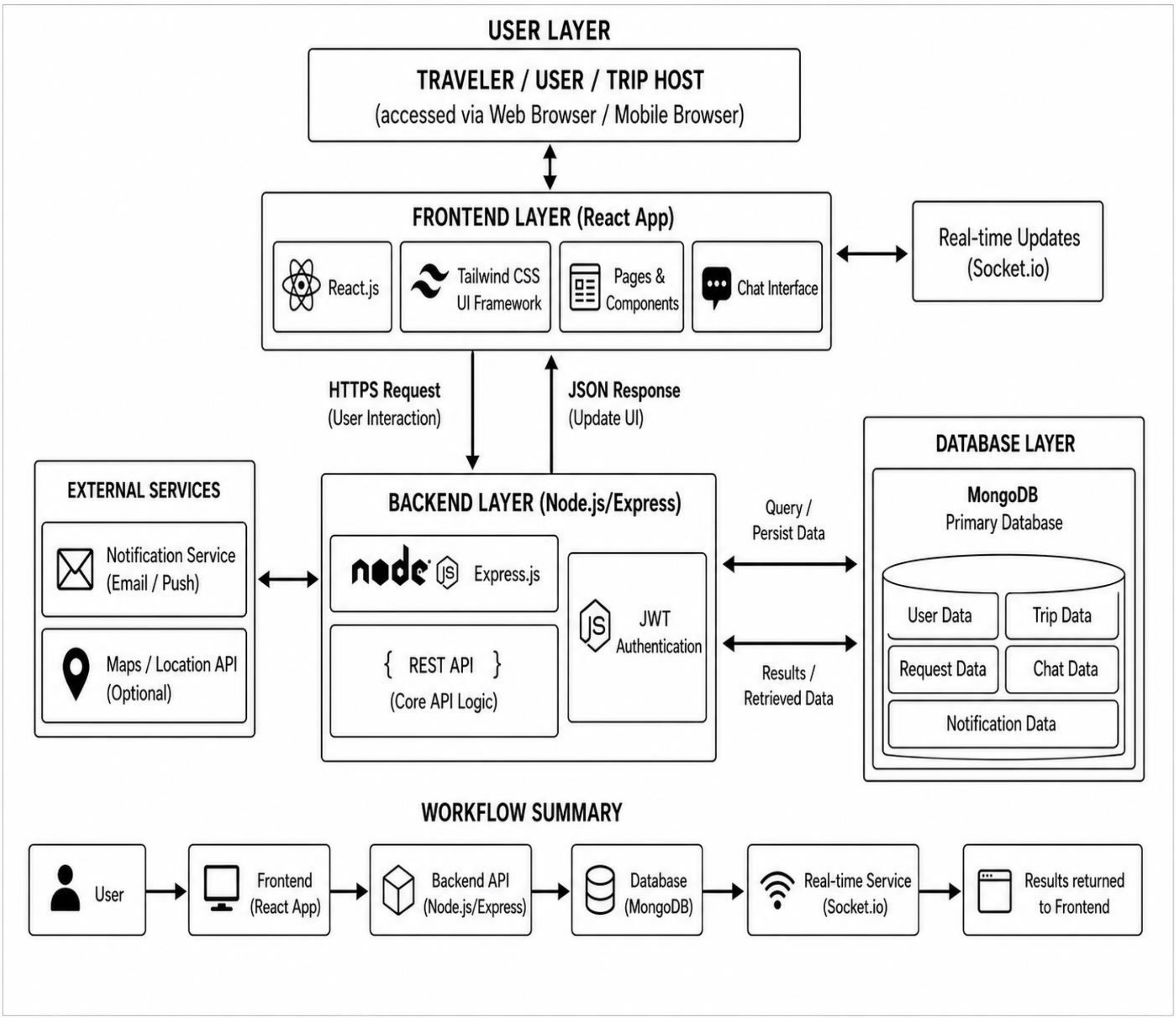

Fig: 3.2

3.3) System Flowchart Diagram:

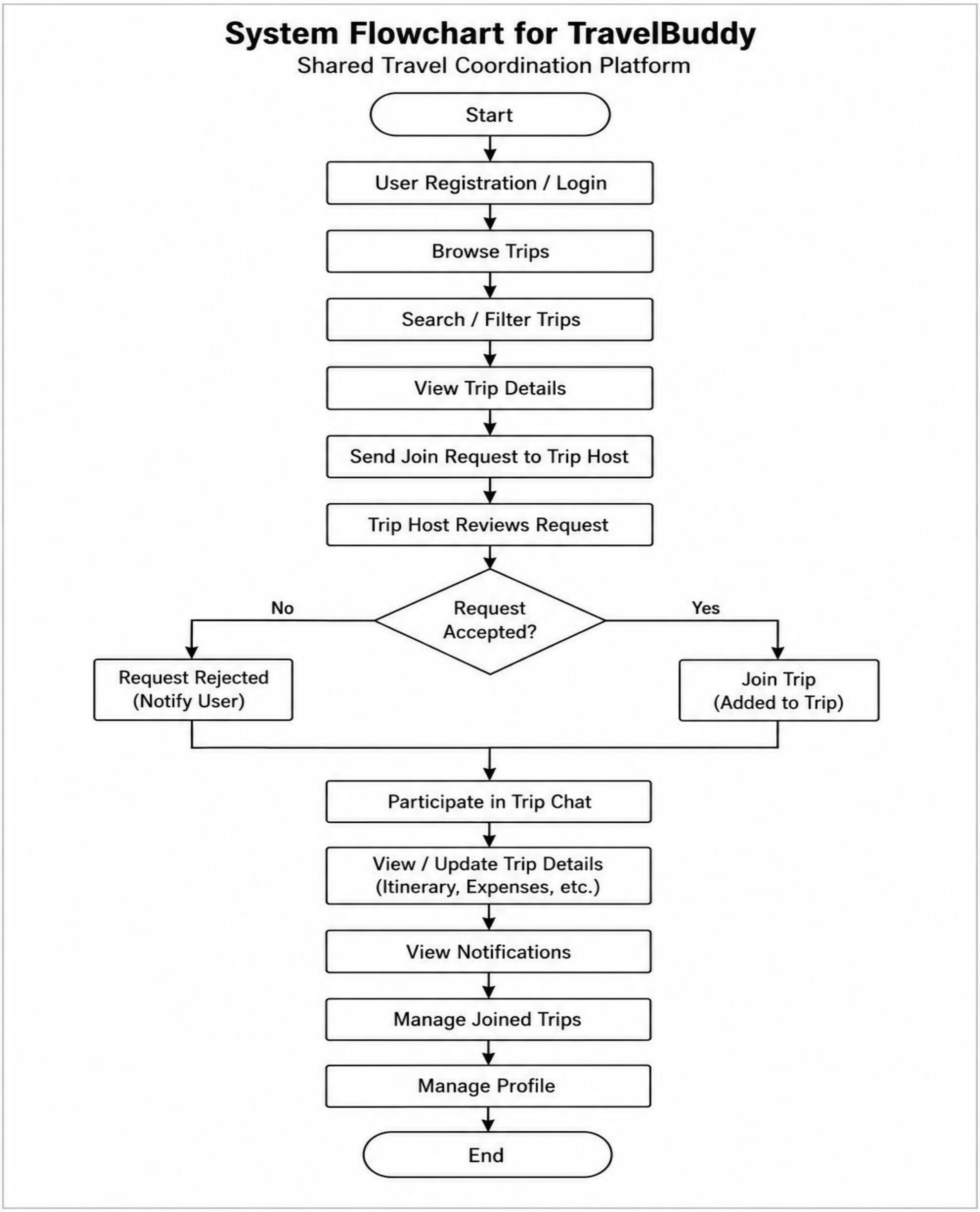

Fig: 3.3

3.4) UML Class Diagram:

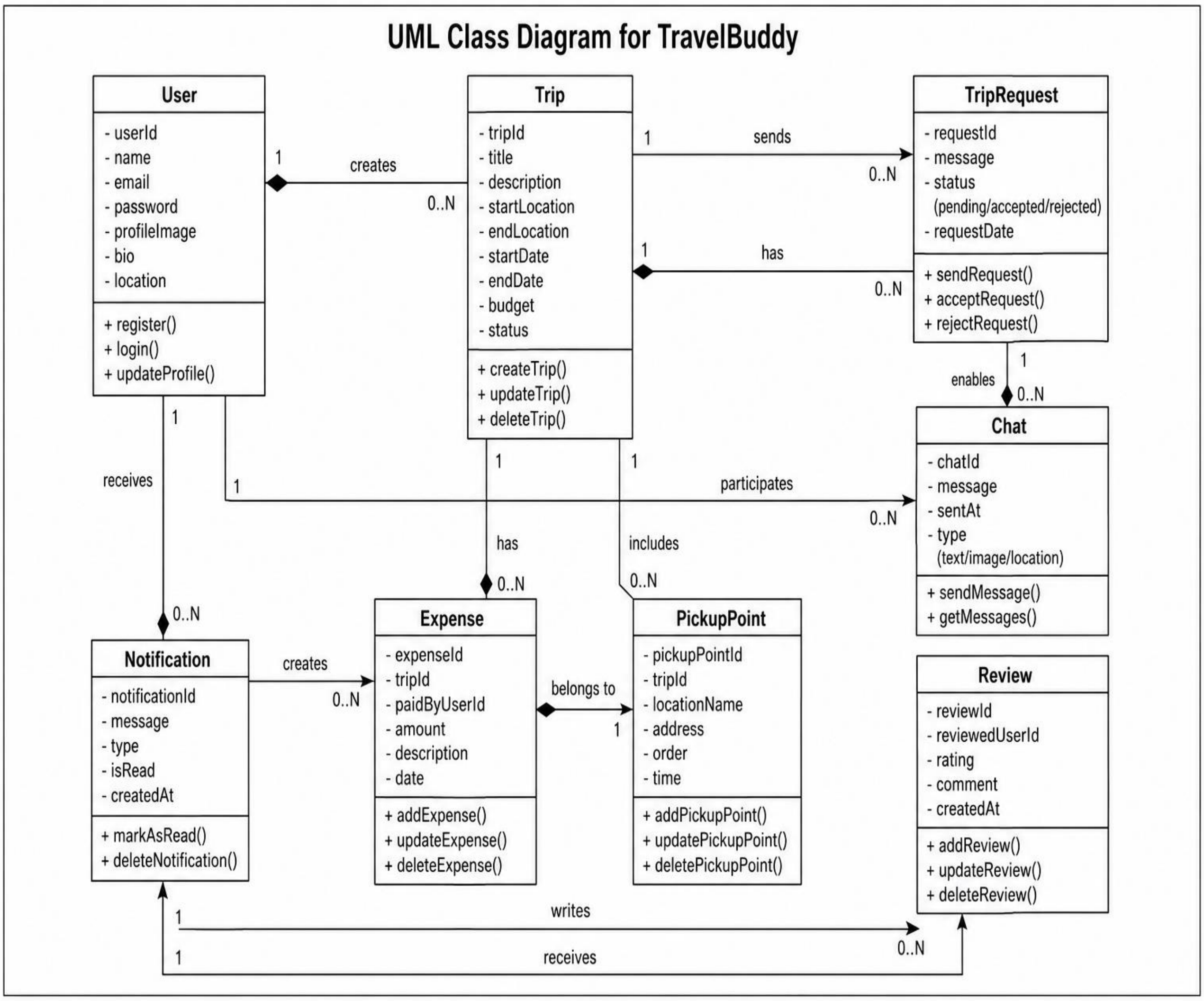

Fig: 3.4

4. Implementation and Results

The **TravelBuddy** platform was successfully implemented using modern web development technologies.

4.1 Implementation

The system includes several modules:

- **User Authentication:** Secure login and registration are implemented using JWT authentication.
- **Trip Creation and Management**: Users can create, update, delete, and manage travel plans easily.
- **Traveler Connection System:** Users can explore trips and connect with travelers having similar destinations and interests.
- **Real-Time Chat System:** Travelers can communicate instantly using Socket.io-based real-time chat.
- **Trip Request Management:** Users can send, accept, or reject trip join requests.
- **Multiple Pickup Points:** The system allows travelers to join trips from convenient pickup locations.
- **Expense Sharing Module:** Users can divide and manage trip expenses efficiently.
- **Notification System:** The platform provides real-time travel updates and notifications.
- **Dashboard:** The dashboard displays trip activities, joined trips, notifications, requests, and chat updates.

4.2 Results

The system successfully demonstrates the integration of modern web technologies with travel planning and traveler interaction features.

The platform allows travelers to manage trips, connect with other travelers, and communicate in a more efficient and organized way. Screenshots of the working system include the login page, dashboard, trip creation interface, traveler connection module, real-time chat feature, trip request management page, and expense sharing module.

These results confirm that **TravelBuddy** provides a functional, interactive, and user-friendly travel planning.

5. Coding

5.1 App.tsx :

```
const queryClient = new QueryClient();
const App = () => (
  <BrowserRouter>
    <UserProvider>
      <QueryClientProvider client={queryClient}>
        <TooltipProvider>
          <ScrollToHash />
          <Toaster />
          <Sonner />
          <ToastContainer />
          <Routes>

            <Route path="/" element={<Index />} />
            <Route path="/auth" element={<Auth />} />
            <Route path='/user/trips' element={<Trip />} />
            <Route path="/user/profile" element={<Profile />} />
            <Route path="/user/my-trips" element={<MyTrip />} />
            <Route path="/trips/request/:requestId" element={<TripRequest />} />
            <Route path='user/notifications' element={<Allnotification />} />
            <Route path="/user/my-trips/track/:tripId" element={<TripTraking />} />

            <Route path="*" element={<NotFound />} />
          </Routes>
            <ChatWidget />
        </TooltipProvider>
      </QueryClientProvider>
    </UserProvider>
  </BrowserRouter>
);

export default App;
```

5.2 authRouts.tsx

```
function Profile() {
  const { user } = useUser();

  const [isEditMode, setIsEditMode] = useState(false);
  const [editData, setEditData] = useState({
    name: user?.name || "",
    phone: user?.phone || "",
    bio: user?.bio || "",
    avatar: user?.avatar || "",
  });
  useEffect(() => {
    if (user) {
      setEditData({
        name: user.name || "",
        phone: user.phone || "",
        bio: user.bio || "",
        avatar: user.avatar || "",
      });
    }
  }, [user]);

  if (!user) {
    return (
      <div className="flex justify-center items-center h-screen">
        <div className="animate-pulse text-gray-400 text-lg">
          Loading profile...
        </div>
      </div>
    );
```

5.3 index.tsx

```
const express = require('express');
const cors = require('cors');
const dotenv = require('dotenv');
const morgan = require('morgan');
const http = require('http');
const { Server } = require('socket.io');

require('./cron/tripStatusUpdater')

dotenv.config();

const app = express();
const PORT = 3000;
const api_version = "v1";

const UserAuthRoute = require('./Routes/UserRoutes/auth');
const TripRoute = require('./Routes/UserRoutes/trip');
const connectDb = require('./Configs/db');

app.use(express.json());
app.use(express.urlencoded({ extended: true }));
app.use(cors());
app.use(morgan('dev'));

connectDb();

app.use(`/api/${api_version}/user`, UserAuthRoute);
app.use(`/api/${api_version}/trip`, TripRoute);

app.get('/', (req, res) => {
  res.send('Server is up and running');
});

const server = http.createServer(app);
```

6. Prototype Photographs for User Side

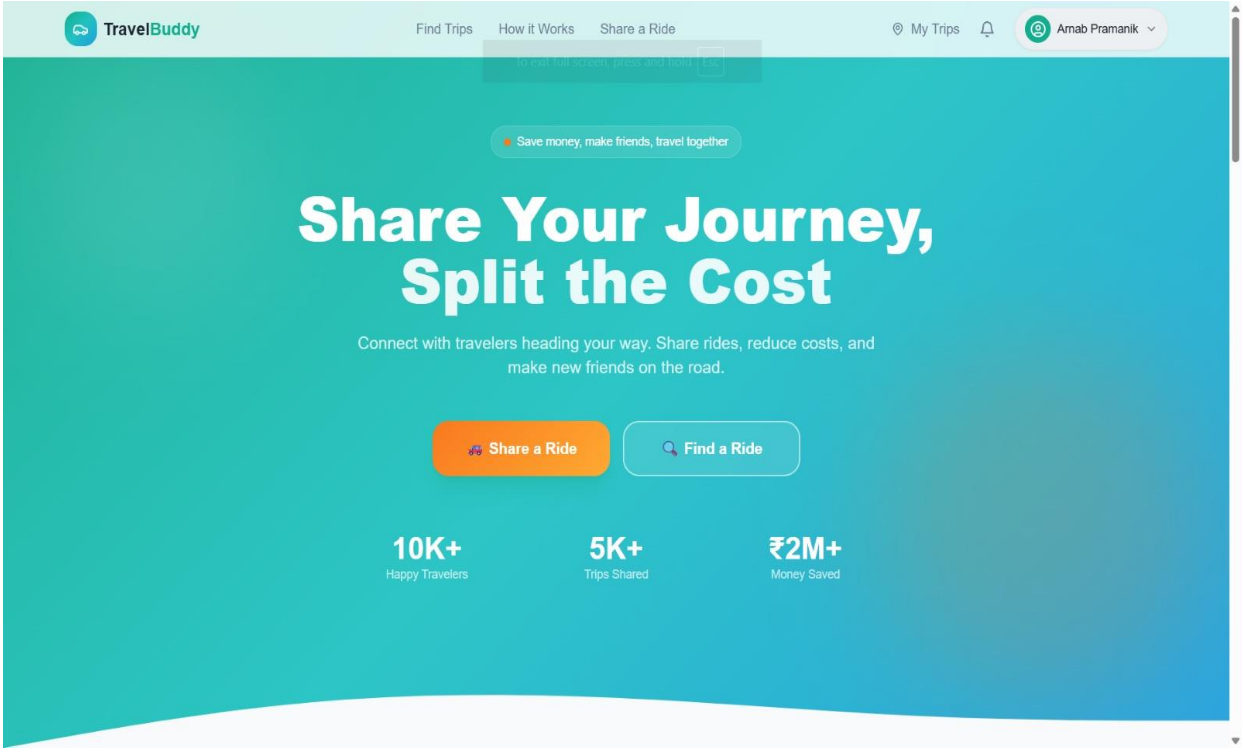

Fig: 6.1

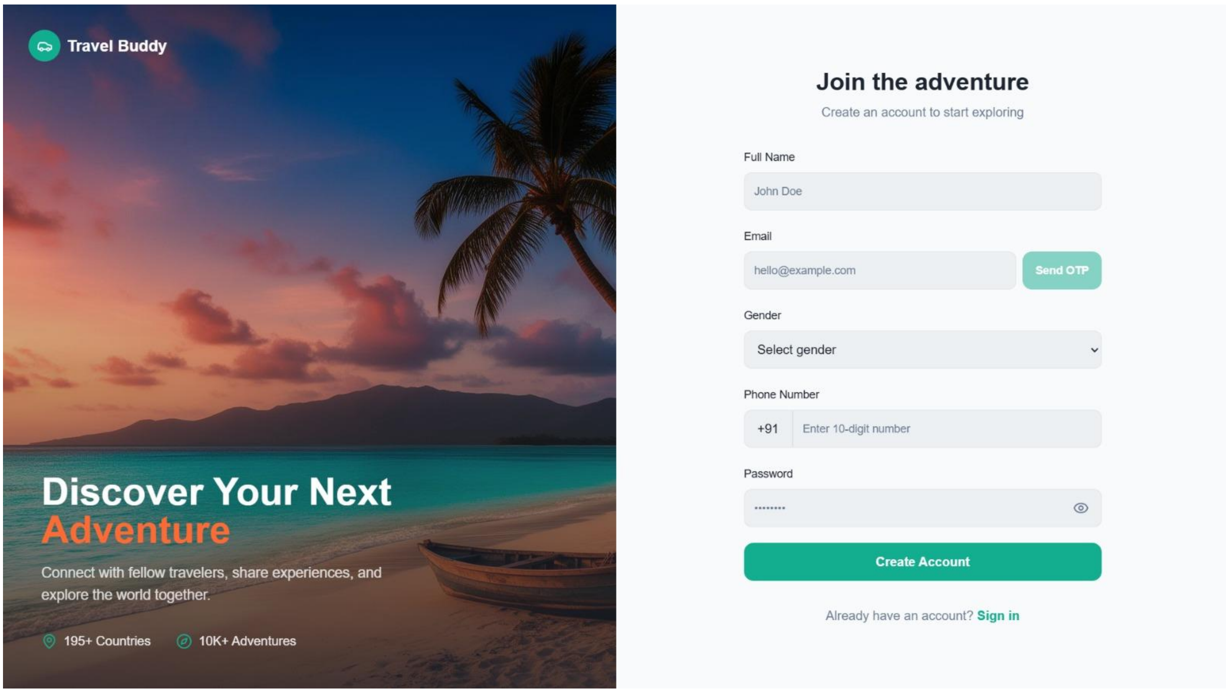

Fig: 6.2

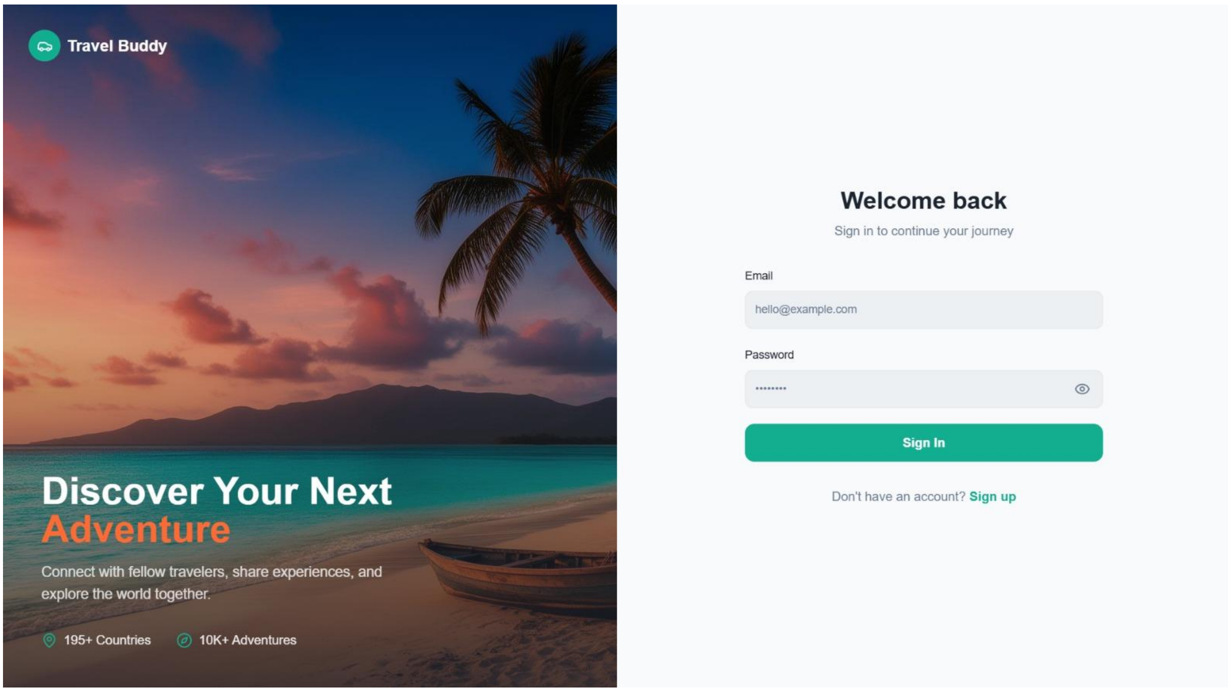
Travel Buddy
Discover Your Next
Adventure
Connect with fellow travelers, share experiences, and explore the world together.
195+ Countries
10K+ Adventures
Welcome back
Sign in to continue your journey
Email
hello@example.com
Password
Sign In
Don't have an account? Sign up

Fig: 6.3

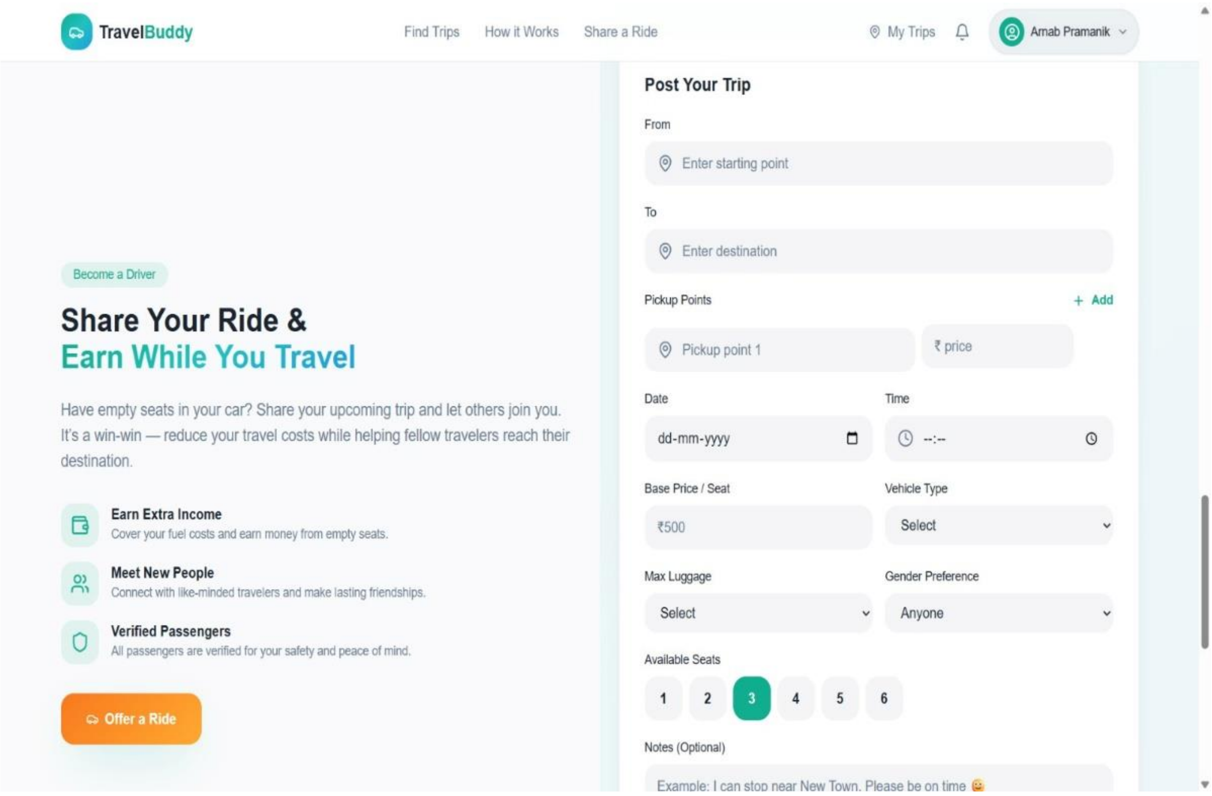
TravelBuddy
Find Trips
How it Works
Share a Ride
My Trips
Arnab Pramanik
Become a Driver
Share Your Ride &
Earn While You Travel
Have empty seats in your car? Share your upcoming trip and let others join you. It's a win-win — reduce your travel costs while helping fellow travelers reach their destination.
Earn Extra Income
Cover your fuel costs and earn money from empty seats.
Meet New People
Connect with like-minded travelers and make lasting friendships.
Verified Passengers
All passengers are verified for your safety and peace of mind.
Offer a Ride
Post Your Trip
From
Enter starting point
To
Enter destination
Pickup Points
+ Add
Pickup point 1
₹ price
Date
dd-mm-yyyy
Time
--:--
Base Price / Seat
₹500
Vehicle Type
Select
Max Luggage
Select
Gender Preference
Anyone
Available Seats
1
2
3
4
5
6
Notes (Optional)
Example: I can stop near New Town. Please be on time

Fig: 6.4

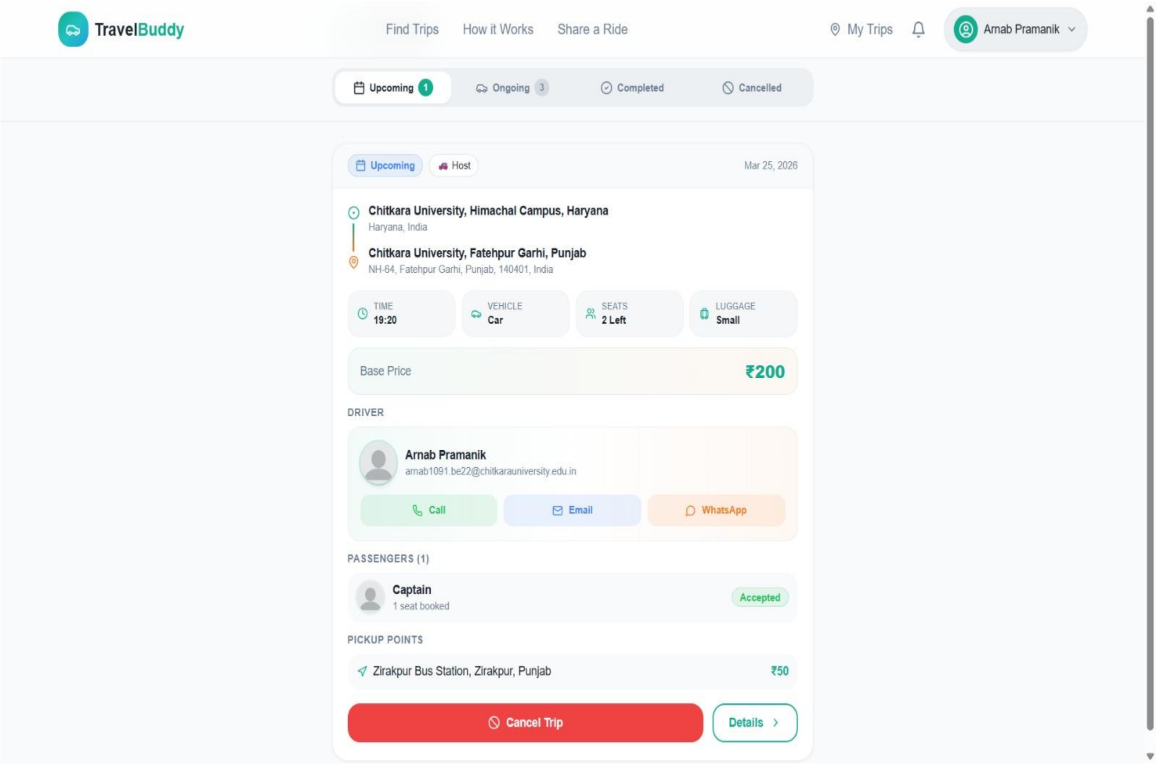
TravelBuddy
Find Trips
How it Works
Share a Ride
My Trips
Arnab Pramanik
Upcoming 1
Ongoing 3
Completed
Cancelled
Upcoming
Host
Mar 25, 2026
Chitkara University, Himachal Campus, Haryana
Haryana, India
Chitkara University, Fatehpur Garhi, Punjab
NH-64, Fatehpur Garhi, Punjab, 140401, India
TIME 19:20
VEHICLE Car
SEATS 2 Left
LUGGAGE Small
Base Price
₹200
DRIVER
Arnab Pramanik
arnab1091.be22@chitkarauniversity.edu.in
Call
Email
WhatsApp
PASSENGERS (1)
Captain
1 seat booked
Accepted
PICKUP POINTS
Zirakpur Bus Station, Zirakpur, Punjab
₹50
Cancel Trip
Details

Fig: 6.5

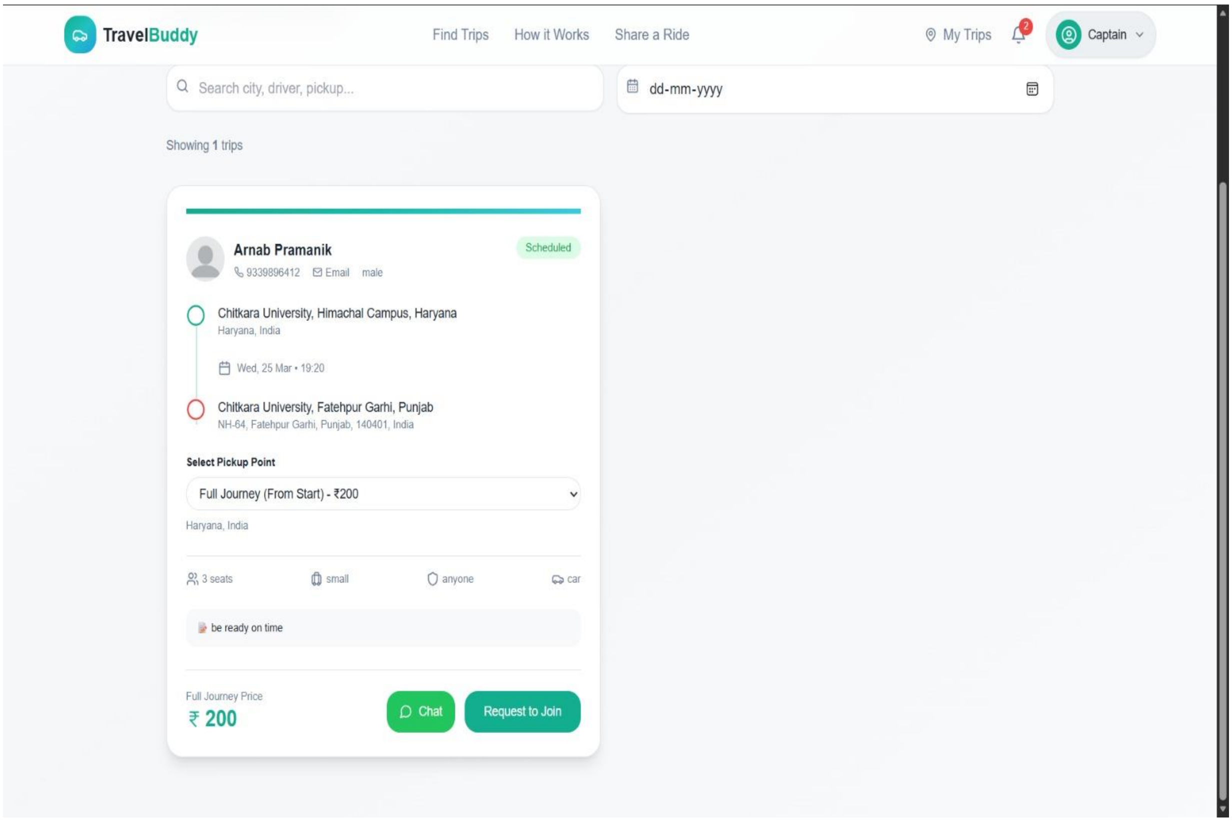
TravelBuddy
Find Trips
How it Works
Share a Ride
My Trips
Captain
Search city, driver, pickup...
dd-mm-yyyy
Showing 1 trips
Arnab Pramanik
Scheduled
9339896412
Email
male
Chitkara University, Himachal Campus, Haryana
Haryana, India
Wed, 25 Mar • 19:20
Chitkara University, Fatehpur Garhi, Punjab
NH-64, Fatehpur Garhi, Punjab, 140401, India
Select Pickup Point
Full Journey (From Start) - ₹200
Haryana, India
3 seats
small
anyone
car
be ready on time
Full Journey Price
₹ 200
Chat
Request to Join

Fig: 6.6

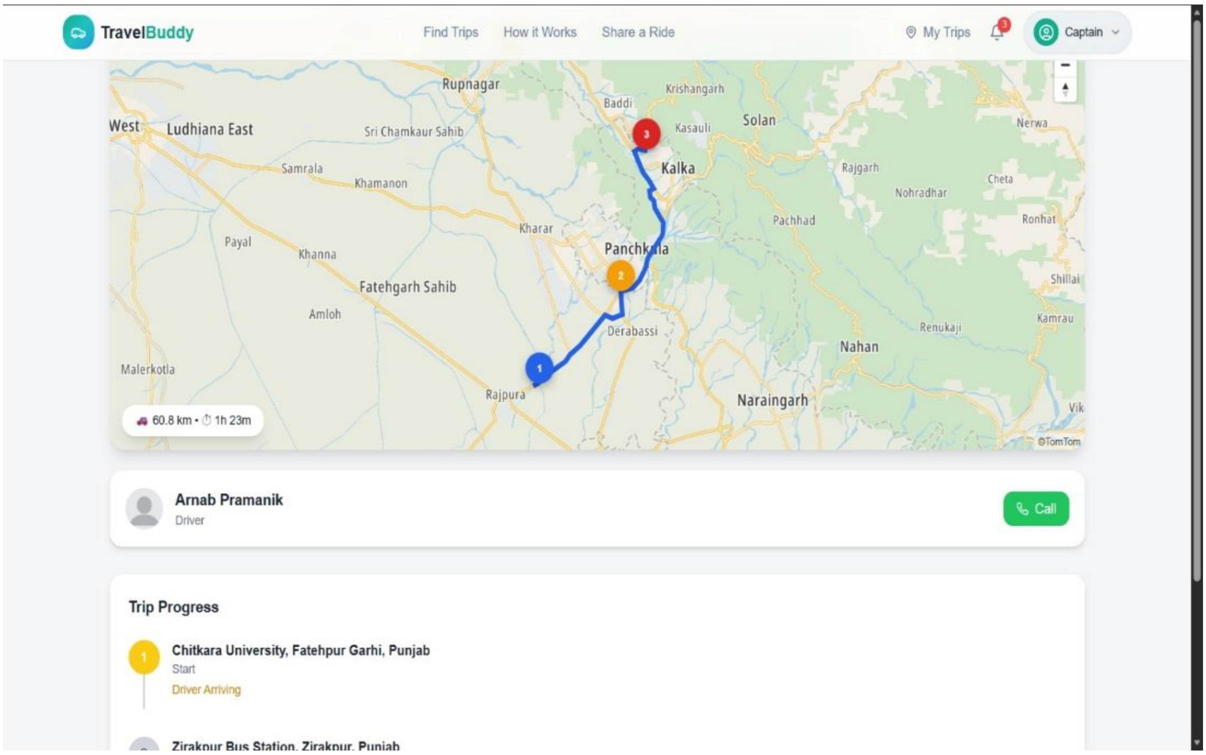
TravelBuddy
Find Trips
How it Works
Share a Ride
My Trips
Captain
Rupnagar
Baddi
Krishangarh
Solan
West
Ludhiana East
Sri Chamkaur Sahib
Kasauli
Nerwa
Samrala
Kalka
Rajgarh
Khamanon
Nohradhar
Cheta
Kharar
Pachhad
Ronhat
Payal
Khanna
Fatehgarh Sahib
Shillai
Amloh
Derabassi
Renukaji
Kamrau
Nahan
Malerkotla
Rajpura
Naraingarh
60.8 km • 1h 23m
©TomTom
Arnab Pramanik
Driver
Call
Trip Progress
Chitkara University, Fatehpur Garhi, Punjab
Start
Driver Arriving
Zirakpur Bus Station, Zirakpur, Punjab

Fig: 6.7

7. Conclusion and Future Work

- ## Conclusion

The **TravelBuddy** project successfully demonstrates the use of modern web technologies in travel planning and traveler coordination systems. By integrating trip management with features such as traveler connection, real-time chat, notifications, expense sharing, and multiple pickup points, the system provides a more interactive and organized travel experience for users.

The platform helps travelers simplify trip planning, connect with suitable travel companions, and manage travel activities more efficiently. The combination of modern web technologies and real-time communication services enables **TravelBuddy** to provide a smart, user-friendly, and reliable travel management solution.

7.2 Future Work

Several improvements can be implemented in future versions of the system.

Possible future enhancements include:

- Integration of live maps and GPS tracking for better travel navigation.
- Development of a mobile application for improved accessibility and convenience.
- AI-based travel recommendations based on user interests and travel history.
- Advanced expense management and automatic cost calculation features.
- Integration of online payment systems for trip bookings and expense sharing..

These improvements can further enhance the effectiveness and usability of the **TravelBuddy** platform.

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2. https://nodejs.org/en/docs – Official Node.js documentation for backend development.

3. https://expressjs.com – Express.js framework documentation for building web APIs.

4. https://tailwindcss.com/docs – Tailwind CSS documentation for responsive UI design.

5. https://ai.google.dev/ – Google Gemini AI documentation for implementing AI features.

6. https://developer.mozilla.org – MDN Web Docs for JavaScript and web development references.

7. https://vercel.com/docs – Vercel deployment documentation for hosting web applications.

8.2 Books

- Learning React by Kirupa Chinnathambi, Addison-Wesley (2nd Edition) – Provides guidance on building modern web applications using React.

- JavaScript: The Good Parts by Douglas Crockford, O'Reilly Media – Explains important concepts of JavaScript used in frontend and backend development.

- Web Development with Node and Express by Ethan Brown, O'Reilly Media – Describes backend development using Node.js and Express.

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